PROJECT TECHNICAL DOCUMENTATION

BioSpecInfo

Web platform for computational chemistry, physics and mathematics

Software Bill of Materials (SBOM)

DOCUMENT CONTROL

Identifier	`BSI-DOC-07-EN`
Title	Software Bill of Materials (SBOM)
Software version	**bsi-v248**
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Document generated from the versioned source in the project repository. The version and revision shown here and in the footer of every page identify exactly the state of the software described.

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Software Bill of Materials (SBOM)

BioSpecInfo · document generated by `tools/genera-sbom.js`.

Application version	`bsi-v248`
Commit	`cd8df42654e3f9738a1e7582ddec35a630c3012f`
Generated (UTC)	`2026-11-11T09:09:35.187Z`
Machine-readable format	`../evidence/sbom.cdx.json` — CycloneDX 1.5

> The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) RDKit / Greg Landrum	`RDKit_minimal.js`	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) RDKit / Greg Landrum	`RDKit_minimal.wasm`	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js mrdoob e collaboratori	`three.min.js`	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass mrdoob e collaboratori	`three_bloom.js`	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader mrdoob e collaboratori	`gltf_loader.js`	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js University of Pittsburgh / David Koes	`3Dmol-min.js`	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).

Component	Version/File	Licence	Size	Role
SmilesDrawer Daniel Probst	`smiles-drawer.min.js`	MIT	246 kB	2D structure drawing from SMILES.
PDF.js (build legacy) Mozilla Foundation	`vendor/pdfjs/pdf.min.mjs`	Apache-2.0	511 kB	Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet.
PDF.js — worker Mozilla Foundation	`vendor/pdfjs/pdf.worker.min.mjs`	Apache-2.0	1290 kB	The PDF decoding work, off the main thread: a complex page does not freeze the interface.
sql.js (JS glue) sql.js / SQLite	`lib/sql-wasm.js`	MIT	45 kB	In-browser SQLite database for the data exercises.
SQLite (WebAssembly) SQLite Consortium	`lib/sql-wasm.wasm`	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) CERN Open Data Portal	`lib/dimuon.js`	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.
Modello 3D «Noir» (Skyfox13, Sketchfab) Skyfox13 — https://sketchfab.com/podderkuntal25	`models/spectra-noir.glb`	CC-BY-4.0	8154 kB	Spectra's face in the Spectra Live console. Modified: textures recompressed, tangents removed, face in dark grey metal, mouth and head animated. Credited in the console.

Digests

File	SHA-256
`RDKit_minimal.js`	`58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9`
`RDKit_minimal.wasm`	`e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8`

File	SHA-256
three.min.js	9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2
three_bloom.js	42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090
gltf_loader.js	5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6
3Dmol-min.js	06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff
smiles-drawer.min.js	955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2
vendor/pdfjs/pdf.min.mjs	bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259
vendor/pdfjs/pdf.worker.min.mjs	145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddaaec6
lib/sql-wasm.js	77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f
lib/sql-wasm.wasm	438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed
lib/dimuon.js	61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d
models/spectra-noir.glb	b781b54819d1d2aabf751788a813d8da830dabaddb913085352b572140903479

2. Code written for this project

File	Size	Role
index.html	4702 kB	Main application: interface, teaching sections, chemical data.
bsi-ai-hub.js	446 kB	«Spectra» AI agent: agentic loop, tools, provider management.
bsi-spettri.js	36 kB	IR/NMR spectral prediction engine over the molecular graph.
bsi-cheminfo.js	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
bsi-pretsch.js	37 kB	The NMR estimation tables transcribed in full from Pretsch–Bühlmann–Badertscher, 4th ed.: ^{13}C and ^{1}H benzenes, ethylenes, alkanes, alkynes, ^{13}C aliphatics, steric corrections, ^{1}J(C,H). Numbers only, no code.
bsi-nmr.js	57 kB	Per-atom ^{1}H and ^{13}C NMR prediction: applies the bsi-pretsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
bsi-geom3d.js	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.
bsi-fisica-comune.js	33 kB	The shared home of the physics sections: header, tabs, sliders, canvas charts, language.
bsi-fis-schrodinger.js	38 kB	1D Schrödinger equation: finite differences, Sturm bisection, inverse iteration, Crank-Nicolson, Morse for real molecules.
bsi-fis-hartreefock.js	48 kB	Restricted Hartree-Fock in the STO-3G basis: McMurchie-Davidson integrals, Boys function, SCF with DIIS, molecular orbitals.

File	Size	Role
`bsi-fis-rotovib.js`	34 kB	IR and Raman rovibrational spectra of diatomics from molecular constants; combination differences.
`bsi-fis-termostat.js`	11 kB	Statistical thermodynamics: partition functions, entropy, heat capacity, Boltzmann populations.
`bsi-fis-dinamica.js`	26 kB	2D Lennard-Jones molecular dynamics: velocity Verlet, thermostat, virial, radial distribution.
`bsi-fis-matematica.js`	24 kB	Mathematics for chemistry: FFT of the FID, Hückel, Runge-Kutta for kinetics, least squares with errors.
`bsi-fis-cristalli.js`	44 kB	Crystallography: Bravais lattices, Miller indices, powder pattern with structure factor and absences.
`bsi-fis-nucleare.js`	14 kB	Radiochemistry: Bateman equations, radiometric dating, Weizsäcker semi-empirical formula.
`bsi-mat-cas.js`	67 kB	Computer-algebra engine: parser, exact fractions, canonical form, derivatives, verified antiderivatives, truncated series, limits.
`bsi-mat-basi.js`	26 kB	Mathematics basics: significant figures, logarithms and pH, equations with working, practice.
`bsi-mat-funzioni.js`	17 kB	Function analysis: domain, zeros, extrema, inflections, asymptotes; families and transformations.
`bsi-mat-calcolo.js`	34 kB	Derivatives with rules, integrals and Riemann sums, Taylor polynomials, limits.
`bsi-mat-lineare.js`	54 kB	Exact-fraction linear algebra: step-by-step Gauss-Jordan, determinant, inverse, eigenvalues, balancing as a null space.
`bsi-mat-ode.js`	38 kB	Differential equations: direction field, Euler/Heun/RK4, phase plane and equilibria from the Jacobian.
`bsi-mat-multi.js`	38 kB	Functions of several variables: surfaces, critical points from the Hessian, Maxwell relations, exact differentials.
`bsi-mat-complessi.js`	37 kB	Complex numbers (Argand plane, roots, Euler) and Fourier series.
`bsi-mat-risolutore.js`	110 kB	Mathematics exam solver: step-by-step working for 16 exercise types.
`bsi-mat-esame.js`	56 kB	Istituzioni di Matematiche I and II exam: syllabus, solver, solved papers, simulation.
`bsi-fis-esame.js`	135 kB	General Physics I and II exam: syllabus, 48 problem models, solved papers.
`bsi-percorsi.js`	9 kB	Study paths from high school to PhD for chemistry, physics and mathematics.
`bsi-mat-probabilita.js`	34 kB	Probability in motion: Galton, central limit, Monte Carlo, Bayes.
`bsi-mat-visiva.js`	24 kB	Visual mathematics: linear maps, Fourier epicycles, fractals, logistic map.

File	Size	Role
`bsi-fis-em.js`	44 kB	Electromagnetism in motion: charges, induction, generator, EM waves.
`bsi-fis-suono.js`	18 kB	Sound and acoustics with Web Audio: harmonics, timbre, Doppler, decibels.
`bsi-fis-solido.js`	32 kB	Semiconductors: Fermi-Dirac, doping, p-n junction, LEDs.
`bsi-chi-distillazione.js`	17 kB	Distillation: Antoine, van Laar, McCabe-Thiele, Rayleigh.
`bsi-chi-voltammetria.js`	12 kB	Simulated cyclic voltammetry: diffusion and Butler-Volmer.
`bsi-mat-numeri.js`	16 kB	Numbers and cryptography: sieve, Euclid, modular arithmetic, RSA, prime distribution.
`bsi-mat-grafi.js`	22 kB	Graphs and optimisation: Dijkstra, Kruskal, gradient descent, linear programming.
`bsi-mat-serie.js`	32 kB	Sequences and series: ε–N limit, tests, rearrangement, radius of convergence.
`bsi-mat-variazioni.js`	51 kB	Calculus of variations: brachistochrone, catenary, catenoid, Fermat.
`bsi-mat-topologia.js`	57 kB	Topology: Euler characteristic, surfaces, fundamental polygons, knots.
`bsi-mat-algebra.js`	45 kB	Algebraic structures: dihedral groups, permutations, finite fields, Lagrange.
`bsi-mat-giochi.js`	36 kB	Game theory: Nash, iterated dilemma, minimax, Nim.
`bsi-mat-logica.js`	46 kB	Logic and computability: truth tables, circuits, Turing machine, sets.
`bsi-mat-inferenza.js`	39 kB	Statistical inference: intervals, tests and power, regression, bootstrap.
`bsi-mat-noneuclidea.js`	66 kB	Non-Euclidean geometries: sphere, Poincaré disk, curvature and parallel transport.
`bsi-fis-gravita.js`	35 kB	Gravitation: N bodies with Verlet, Lagrange points, gravity assist.
`bsi-fis-calore.js`	17 kB	Heat and diffusion: 1D and 2D heat equation, Newton cooling, walls.
`bsi-fis-ising.js`	26 kB	Ising model: Metropolis, phase transition, hysteresis.
`bsi-fis-particelle.js`	59 kB	Particle physics: Standard Model, conservation, invariant mass, lifetimes.
`bsi-fis-analitica.js`	40 kB	Analytical mechanics: phase space, double pendulum, Noether, Liouville.

File	Size	Role
`bsi-fis-laser.js`	34 kB	Lasers and quantum optics: threshold, cavity modes, Rabi, Mach–Zehnder.
`bsi-fis-clima.js`	34 kB	Climate physics: radiative balance, CO2 forcing, ice-albedo, Coriolis.
`bsi-chi-fasi.js`	38 kB	Phase diagrams: water and CO2 (IAPWS, Span-Wagner), eutectic, Clapeyron.
`bsi-chi-pes.js`	33 kB	LEPS potential energy surface for H + H2 and trajectories.
`bsi-articoli-nobel-fisica.js`	118 kB	Topic explanations of the Nobel Prizes in Physics (no data).
`bsi-articoli-premi-matematica.js`	123 kB	Topic explanations of the Abel Prize and Fields Medal (no data).
`bsi-materie.js`	11 kB	Subject switch buttons: chemistry, physics, mathematics.
`bsi-fis-astro.js`	36 kB	Astrophysics: H-R diagram, black body, solar system, exoplanets, cosmology.
`bsi-fis-relativita.js`	30 kB	Special relativity: Minkowski, twins, dynamics.
`bsi-materie-app.js`	15 kB	Three apps in one: the chosen subject adapts the ✨ menu, Spectra, Study and Utility.
`bsi-attrezzi.js`	23 kB	Quick tools of the ✨ menu for physics and mathematics: calculator, derivative, graph, constant, units, formula, glossary, quiz.
`bsi-studio.js`	66 kB	Study-kit engine: summaries, quizzes, Leitner flashcards, formula sheet, glossary.
`bsi-studio-fisica.js`	306 kB	Physics study-kit data (no code).
`bsi-studio-matematica.js`	290 kB	Mathematics study-kit data (no code).
`bsi-risorse-materie.js`	149 kB	Web resources, bibliography and citations for the three subjects; CODATA constants and units; mathematical tables with their proofs.
`bsi-astro-sistemasolare.js`	110 kB	Solar system 3D at any date: Standish JPL elements, Kepler, Earth, Moon and eclipses, Jupiter and Saturn.
`bsi-astro-orbite.js`	69 kB	Orbital mechanics 3D: elements and state, manoeuvres, satellites, Lambert.
`bsi-astro-matcelesti.js`	60 kB	Celestial mathematics: Kepler equation, rotations and quaternions, N-body integrators, periodograms.
`bsi-astro-stelle.js`	63 kB	Life of stars 3D: Lane-Emden, evolution, nucleosynthesis.
`bsi-astro-buchineri.js`	72 kB	Black holes and general relativity: Schwarzschild ray tracing, geodesics, curvature, Hawking.

File	Size	Role
`bsi-astro-ondegrav.js`	55 kB	Gravitational waves: inspiral, chirp, detector, Hulse-Taylor pulsar.
`bsi-astro-galassie.js`	70 kB	Galaxies 3D: Hubble types, rotation curves, collision, Local Group.
`bsi-astro-cosmo.js`	70 kB	Cosmology 3D: Friedmann-Lemaître, distances, CMB, thermal history.
`bsi-astro-cielo.js`	75 kB	Planetarium 3D: the sky from any place and time, coordinates, precession, analemma.
`bsi-astro-esopianeti.js`	57 kB	Exoplanets 3D: habitable zone, transit, radial velocity, direct imaging.
`bsi-fis-spin.js`	55 kB	Spin, qubits and entanglement: Stern-Gerlach, Bloch sphere, CHSH, BB84.
`bsi-fis-statistica.js`	47 kB	Statistical mechanics: Schottky, Maxwell-Boltzmann, Fermi and Bose, Einstein and Debye.
`bsi-fis-nuclei.js`	49 kB	Nuclei and reactions: valley of stability, Q values, tunnelling, 3D nucleus, reactor.
`bsi-fis-materiali.js`	48 kB	Elasticity and materials: tension, beams, Euler, sound in solids.
`bsi-fis-elettronica.js`	48 kB	Electronics: rectifiers, diodes and Zener, common emitter, op-amps, ADC and DAC.
`bsi-mat-edp.js`	75 kB	Partial differential equations: string, membrane, heat, Laplace, Burgers.
`bsi-mat-analisicomplessa.js`	70 kB	Complex analysis: domain colouring, conformal maps, Joukowski, residues.
`bsi-mat-markov.js`	65 kB	Markov chains and stochastic processes: PageRank, gambler's ruin, walks, queues.
`bsi-mat-svd.js`	58 kB	SVD and applied linear algebra: 2×2 and 3×3 SVD, compression, PCA, least squares.
`bsi-mat-codici.js`	60 kB	Codes and elliptic curves: Hamming, CRC, curves over ℝ and F_p, ECDH.
`bsi-spectra-live.js`	113 kB	Spectra Live: voice console, local commands, screen reading, nine agent tools.
`bsi-spectra-androide.js`	45 kB	Spectra's face on the «Noir» model: animated head and mouth, 17 emotions from 11 expressive fields.
`bsi-spectra-mente.js`	17 kB	Spectra's mind: local memory, internal state, thoughts (a simulation of behaviour).
`bsi-spectra-robot.js`	20 kB	Fallback face: android head built in three.js.
`bsi-spectra-volto.js`	16 kB	Last-resort face: SVG vector portrait.

File	Size	Role
`bsi-lezione.js`	33 kB	Guided lesson with Spectra for every section: chapters from the tabs, controls demonstrated, spotlight, check question from the study kit.
`bsi-spectra-voce.js`	12 kB	Spectra's voice: TTS audio (with an OpenAI key) through a Web Audio chain (bass, metal, room) and lip-sync from the audio; falls back to browser synthesis.
`bsi-spectra-corpo.js`	49 kB	Body driver for any humanoid avatar (bones, ARKit blendshapes, black hair, gaze, gestures, walking, sitting) and a rigged test mannequin.
`bsi-spectra-olointerattivo.js`	74 kB	Spectra's interactive hologram: IR/^{1}H/^{13}C/mass/UV spectra of reference molecules or of a written trace with each signal assigned; universe with NASA data, Kepler and vis-viva; maths and physics with the assistant's working; every step explained aloud and in writing.
`bsi-spectra-esplora.js`	42 kB	«⚡ Explore with Spectra»: one section per subject in the style of explorable lessons, ten chapters from high school to advanced PhD with prediction, a recomputing slider, live view in the hologram and full explanation.
`bsi-spectra-spiega.js`	36 kB	«🧠 Explain»: the complete, exhaustive explanation of chemistry, physics and mathematics from high school to advanced PhD, including what is on screen (text, values, tables and an image of spectra, graphs and 3D scenes); without a model it shows what the app knows on its own.
`bsi-spectra-figura.js`	55 kB	Spectra with a body in one module: the body controller (embedded in the app and packed, rig, aligned «Noir» face) used by the studio, and the figure in the AI sections that thinks, speaks and gestures with the answers.
`bsi-spectra-rig.js`	35 kB	In-browser auto-rig for humanoid figures without a skeleton: measuring, Mixamo-named skeleton, skin weights (distance, normal, geodesic), head removed keeping the hood, night-blue recolouring.
`bsi-spectra-studio-lezioni.js`	34 kB	Spectra Studio lessons (predict → test → prove → reveal): the fall, the shape of water, the Basel problem, with verified computations.
`bsi-spectra-creatore.js`	11 kB	Spectra recognises its creator: a secret-phrase check compared as a SHA-256 fingerprint (a deterrent, not access control).
`bsi-aggiorna.js`	5 kB	«Check for updates»: version in use from the service worker, published version from sw.js.
`bsi-fis-fluidi.js`	31 kB	Fluids: Venturi and Bernoulli, viscosity and Stokes, capillarity.
`bsi-mat-vettoriale.js`	43 kB	Vector calculus: fields, Green, gradient.
`bsi-chi-cinetica.js`	31 kB	Complex kinetics: composite and oscillating reactions, Turing patterns.
`bsi-fis-meccanica.js`	37 kB	Mechanics in motion: projectile with air, collisions, real pendulum, coupled oscillators.
`bsi-fis-onde.js`	33 kB	Waves and optics: superposition, lenses and mirrors, diffraction.

File	Size	Role
`bsi-fis-circuiti.js`	17 kB	Circuits: RLC transients, alternating current, filters and Bode.
`bsi-fis-termo.js`	44 kB	Thermodynamics in motion: cycles, particle gas, Brownian motion.
`bsi-mat-numerici.js`	24 kB	Numerical methods: roots, integration, ODEs, interpolation with measured orders.
`bsi-mat-geometria.js`	24 kB	Analytic geometry and vectors: vectors, lines and planes in 3D, conics.
`bsi-chi-equilibrio.js`	27 kB	Chemical equilibrium and Le Chatelier: kinetic simulator and K(T).
`bsi-lab3d.js`	63 kB	3D Laboratory: surfaces,
`bsi-vista-molecole.js`	8 kB	Quality settings and control bar for every 3Dmol molecular viewer.
`bsi-assistente.js`	71 kB	The three macro-areas Chemistry, Physics and Mathematics: natural-language assistant with full working and inverse problem, tool catalogue, levels.
`bsi-chi-esame.js`	98 kB	Chemistry exam from basic to expert: 40 problem models, solved papers.
`bsi-fis-elettrodi.js`	14 kB	Electrode kinetics (Butler-Volmer, Tafel) and transition-state theory (Eyring).
`bsi-fis-superfici.js`	18 kB	Photophysics (Jablonski, Stern-Volmer), adsorption isotherms (Langmuir, Freundlich, BET), polymers.
`bsi-mat-statistica.js`	27 kB	Laboratory statistics: t, F, Grubbs, Dixon, uncertainty propagation, calibration line with LOD and LOQ.
`bsi-nmr2d.js`	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
`bsi-moldraw.js`	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
`bsi-documento.js`	23 kB	Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read.
`bsi-quesito.js`	19 kB	Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js.
`bsi-spettrolettore.js`	104 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
`bsi-mol3d.js`	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.
`bsi-lingue.js`	258 kB	Internationalisation engine: fourteen languages, writing direction included.
`bsi-molingue.js`	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.

File	Size	Role
`bsi-pannelli-lingua.js`	66 kB	Panels for languages and chemical languages.
`sw.js`	9 kB	Service Worker: offline operation and network strategy.
`rdkit_lab.html`	260 kB	Cheminformatics laboratory.
`astro.html`	2660 kB	Astrochemistry module.
`chimorga.html`	194 kB	Organic chemistry module.
`proxy/spectra-proxy.js`	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
`index.html`	`7935a1667d786d39240ff2cf4273a2aed5cdf6ed0b58c16f25755893932cf33b`
`bsi-ai-hub.js`	`c2db3c146724e49d9694cb3b3476301274475200290 8c289c79bab3259265f2b`
`bsi-spettri.js`	`36911b64fd173917f9151c8ef82fc1c6f903eb93205fc851f20534a46d958921`
`bsi-cheminfo.js`	`59a3111b3b9436add5376fde1be6f90ded3dc08c095cebdc04da50c7f8268bbc`
`bsi-pretsch.js`	`27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3`
`bsi-nmr.js`	`881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72`
`bsi-geom3d.js`	`a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054`
`bsi-fisica-comune.js`	`5fe08ef43a14a93e09caef439a07f0742221bdf1d395ce1c50dfed9e90cc73bb`
`bsi-fis-schrodinger.js`	`9954f7b14d9c84a2539480c2e61b468f2c37c72d03b3a6147d414c9870480d9b`
`bsi-fis-hartreefock.js`	`541b5340edbb392f5f85d473f430a586282a848e0dc432b752ab898407582c84`
`bsi-fis-rotovib.js`	`0ce3ad61451a86506e9e653776d73b9e2cd21986c84a00e65db0fb6290ebbdde`
`bsi-fis-termostat.js`	`0bb9f04d7d5b144791764fa4efd9f8e658be5b5d78b40f94e18752c2366589a9`
`bsi-fis-dinamica.js`	`0b1ab11071eab8a1af5def46919bc73aa2dbd9dd58c77cea30aa6dee020f4755`
`bsi-fis-matematica.js`	`616a2d4f338fe1d371f5ba3fd4fe569e94b2d94626a8d323bba8ae58f8550f62`
`bsi-fis-cristalli.js`	`02e2224e77713d071720d6394452a21eb8e9e1c9103871f41193e82f346fe805`
`bsi-fis-nucleare.js`	`8ae148a2f37a40f6902faa4148e44468b4d659b6c7d1230378de92c2703a5874`
`bsi-mat-cas.js`	`eea3a1eef74d31d3fde78b7e4a2795169e8af0f357edc10050eb98d8eaf386ac`
`bsi-mat-basi.js`	`db30e16c58bde3638f13de5c3311a00598628676b7b418dc023b175efa77c214`
`bsi-mat-funzioni.js`	`68f8780d4fb4fcda6f028e4749adef333253efbf270d0caab47def5bdf30c66c`
`bsi-mat-calcolo.js`	`3d5cff0520cf385dfe35af1b3f916a66c5afb4ff60d880f04123ac5fe73a9279`
`bsi-mat-lineare.js`	`a84ca3fb6720da50f474633152b84e03e1d2287e3271cdadca790b735c69d686`

File	SHA-256
bsi-mat-ode.js	bfbb1528e5d025d7492c69503115dc0dfb1c50fbbd6735133306ad1327830163
bsi-mat-multi.js	5a025543a12c66994e8ac08a2705da5761eac8bbcc5a5be471536372762b2e37
bsi-mat-complessi.js	8ad8fff97934e7224d97523decec29bb3b4bc4631dc20cc6ea78a0caec69320b
bsi-mat-risolutore.js	709349e1d81f03506c02b9b05a060d8d3801dc37183c0e3894cc93c0aafe9f0c
bsi-mat-esame.js	de4af253244bd7d610b1a3607297d1678a529504afa23d7123aebe042e1f2c27
bsi-fis-esame.js	c9ffb883e94a07b7cda3777b34d6141bd4c47bb3c9857218a9a37f65f8728947
bsi-percorsi.js	950c686202f2934009827bc300c787063b325528b7d5ab5385a8ab3b6706345c
bsi-mat-probabilita.js	adc628789ec90772c14ce0e61aed74981cf60b261ac15ff6076d1d4b3e8d0bec
bsi-mat-visiva.js	cb624f3e1e6c6df94c287f961d0f21a9d8ec852e8245ff5aeed09a083f2c5b14
bsi-fis-em.js	1c45fb608ccda1e500ab3038998cee4cb9c27b091c0b1d1b4f7af0137bf9a6ab
bsi-fis-suono.js	fead9e5dde961fc56c3a649cce6e8eed23046f327c8392dbfd8de5024cf0a1bf
bsi-fis-solido.js	e185ce2e4f5cb4e1101bbb5d398fa3893a017c8df8fa4a26246291fda60ca2ad
bsi-chi-distillazione.js	8d13592cf7523c1dd159d3e954947a4956e44539308312dc97cd3588207fcfc0
bsi-chi-voltammetria.js	4db1a71e0ac3569b821f6cc5280470453f3a4df5c62b9695a3339367c5aa3e41
bsi-mat-numeri.js	f30d041cd9963342aa476e128e4b29d543e113d3483253377ea486de3d46484f
bsi-mat-grafi.js	4670e67bd25b97d4c7d2b8e72b03ed8ed4f6e6aa5c1c43dd74c56b437c1c8c33
bsi-mat-serie.js	722b1a1b7397b7710689af555d1fd60de73ee405f5061819d9e59d7f4631d060
bsi-mat-variazioni.js	ccb7379618067a0412d27fe95cb19d5823e0c0b739a371afc7cc69f162a5ca7a
bsi-mat-topologia.js	87a8df768b8fa968608b52462df4340484ccaeb7ed3e6a987987a62603ceb8fa
bsi-mat-algebra.js	59d66b5c6be431f49e5808e5c78758a7ca3cc33fba6155e56457f13dec14e237
bsi-mat-giochi.js	d1ae8048acab5a518f1ca9c9ef31e4c9346459c7a0d83ec935733a57465c4b72
bsi-mat-logica.js	26cfc591156a6c232dd40938252ce238374a6a678f08557d3bcfa899a5d0785e
bsi-mat-inferenza.js	4dadc036537e3833e396ab5dfa9968f0ccab398ccc18959ab474e6652d27e43c
bsi-mat-noneuclidea.js	a687dbe43e91cacc5b2287af6e135188363e49644cd870b48db9b4a66d555119
bsi-fis-gravita.js	195d35280eb3484f0374b9e51801e96634c797ba21209fce1bdb7b8aed867e33
bsi-fis-calore.js	81b9c1c60c7ba5a022f2f95e36ac17274ed1e2027d0ba36bcc24f8d6243ad3b8
bsi-fis-ising.js	8ccd29082a4388ce63bdc1e39d9aa5fa92322e378384df0951d98bde6fe79b8c
bsi-fis-particelle.js	4dce551cb464150e0faae0bcddcea834059692187310450 3bd6a2dd10b41140c
bsi-fis-analitica.js	01b4178998372c6db7d18a312c5a5ed1c8c26267bd0b72ae63ad6fbf92ba2895
bsi-fis-laser.js	8f876f1552b50b22643526a2d16a4a8b0c0bbf5b8d9161d325f38efd16c917fa
bsi-fis-clima.js	8e136c02f7559cd22d57ccd857d06f8d6a6a89b2f38790d2782e6a45a7f2c822
bsi-chi-fasi.js	99854760b11db89cf853bb4c2be08d00f018fb510fd6a56aff1c3006e68ed1af
bsi-chi-pes.js	c4e74b909e7987275930b05c0c25cdc0968bcf338ea78a144bc7ff949a821e36
bsi-articoli-nobel-fisica.js	afc22b073e64168605b3d792cd83bcdf7d838e0c80548a0a58caae98456d7d30

File	SHA-256
bsi-articoli-premi-matematica.js	d62f01d66783d7fb558684959abd1cda2f81ee6885c8cdc0879559f2b64cbf43
bsi-materie.js	763063fb3d3c1e43da8066c6febab6309f889a6550e7da1f45ce1d001c27c314
bsi-fis-astro.js	64901258a641c54ee99a862990a7e5d86acc5c2367408bb38db9371a39c7363e
bsi-fis-relativita.js	d90cb347a69e668fde69df4805815938f8da8b9a5f288ce7a16914aa0fec32bd
bsi-materie-app.js	a9d57912a69bf7c599a6c4d16be815be4e654ab424daeca0e12f02c00904911f
bsi-attrezzi.js	dedf9939e96c9544239f5769b68f0547604aff78e3ee3f47906afddcf1e20a38
bsi-studio.js	aa1c97df67465affd4f7d0c96240b0d0ae6dd9d8c8dd6bf3c93abe9cef49e377
bsi-studio-fisica.js	973bab09dad47980f0d411aef35ff0378f2266030c3fcb14e1cefe64eb63531b
bsi-studio-matematica.js	3a2d88a7f9cea52bc5c46bf86018904b2d032320726988759b3e0fb84110c5c7
bsi-risorse-materie.js	03e2fa89af1157985cdab1a08c60153ed57d8c648a5f7b67cd3f5d910c43e417
bsi-astro-sistemasolare.js	d59c52b7135d020d67a8163e9bc4d0cecea2446df93aa66389f0477a2a908ef7
bsi-astro-orbite.js	32c8123ac5a7b39a9a4a816713d0305a9d90ddb8fb714614fcbba3eb3e834c4b
bsi-astro-matcelesti.js	b2ee40c8a53fb25baf42020693539d91dc5278a14436115c69643470d57cda56
bsi-astro-stelle.js	84064a5bdf8fdc1111011201cf97a11a47396acbd3c7442d52e2c5ba74a388a3
bsi-astro-buchineri.js	c75a84cba9be45c5fec1b4f77f4ca79f9d898067006a3c309d951147ddde083c
bsi-astro-ondegrav.js	dcbc8ea2e45f55a463ea344c26e0de8e0aa6a14f6adc73d6b2a2e43aee9736db
bsi-astro-galassie.js	893eee603ec4af64ceaf440f3a22da3dab77a55333911e80e06212459e5f7125
bsi-astro-cosmo.js	356152c5251593c68e8dd66b1182f5ee3f7c5f4f1cb26c722bdd384f8fe66d47
bsi-astro-cielo.js	26b3e33cefe7d1606d0abf8d47be6d7d9b412891ab193d386835d4e40f9fdcf6
bsi-astro-esopianeti.js	5a540d026927fc13f0081f3ba4cbfcb6de4c44056530571c77574e71f9003a49
bsi-fis-spin.js	a24dc05385fbff4826a02f95315ae1d8c57c8ee555da868a08620f24127fe3ad
bsi-fis-statistica.js	b4fd532901c0ed72d0d104c85d686d534cc90b59eacd8eaaed98255cfea553b2
bsi-fis-nuclei.js	3f393cf853d4171beb53153fa2d03da2f526ba40af2b9ad1ce4c4e426678d4db
bsi-fis-materiali.js	b2869b83fa17200da5f3900abf6959214df0dd6f88c4d8d02be88d5a6d832269
bsi-fis-elettronica.js	8620e8ba116f1d3edb3a487a7bf2aab869fbf1afc2230462e251062e7a2bf61d
bsi-mat-edp.js	525fae4f04fcc0ec7884ebba30e47d013c972a5ff77b1088ab4bca41bd04a50b
bsi-mat-analisicomplessa.js	0eb6e5181471af7e85eb3ee42012daddbe1fdac932c0c4c28cf67135875302d8
bsi-mat-markov.js	0e122fac691f1faa2e464214a7d24c9a687b16491c5964b760e0df191b13a35b
bsi-mat-svd.js	ce6d77e921c469c532f9f6de2689f7b2a8c8de656f18ee32b1fc4e1977538931
bsi-mat-codici.js	9aaf44beb1922a952392850679419 99e6ae4424024546c8b38dc2663e5f091b2
bsi-spectra-live.js	3b46a1d764b8804d46d444149e46eea9f0bfa292c2300f8dcbcaefe43b99879f
bsi-spectra-androide.js	36bd9ad900e9da2860ddf95469c48ff81387d6da6ca71672bb52975cebb6bd08
bsi-spectra-mente.js	29a9666fb7278adcd8747060927f29403b53b47ab00e5d2b5dc6e9ccff311436
bsi-spectra-robot.js	227944042a092f360a6f27e2d8221686101fa6f29514c4e56eb4401d7de5b1b7

File	SHA-256
bsi-spectra-volto.js	7061605f2d7e124d2a80ec2f19531b7fa32fed843b60ca3d07843f4926105bf6
bsi-lezione.js	9d254c1cfaaf0c9ea5ec33086e71c665941f352378a14c874e27b3cb7e8f861a
bsi-spectra-voce.js	fa6f3440ffadc9a9a4c2b57821141ea3d2d4ef36dca9f85d92dc21e652dad9a7
bsi-spectra-corpo.js	ae8e5ea95d2a8946b7318d5ac4f4e3646297a4ebeeb30924d8f8c1bd0110a04f
bsi-spectra-olointerattivo.js	19cd15b7c5e74fd97992aba0c13defbd533dbcd386aaf996f9a5621a53bf04b8
bsi-spectra-esplora.js	71b0362cbc8c12442ba5fe9b804bb982b95911afce317684a4cdf418586f7c91
bsi-spectra-spiega.js	789b3560c1372da180a86a486e9ce26f1d471768d39bd39fe1c6f55b85dc8b95
bsi-spectra-figura.js	1c072d7b2f9de5fdf627dc985e200366c1ef383ed41ff088d9380f4fbea6ec98
bsi-spectra-rig.js	fcdd67c468f543a6e98f682d49c45b26173bb987c1aa529cb63d4e3e2a40918b
bsi-spectra-studio-lezioni.js	a7344a2b7a62986827dff100a44676d10daa841d76af97b056b796657a991a9c
bsi-spectra-creatore.js	73954030d3fac2a5b12fd26c92f61a1c8bb986813941d33fb6b154e8cf89bee2
bsi-aggiorna.js	5950d90103a7d603175e4b577e2394bb9a6d006a4ff22e48d4a1cc70661f1b8e
bsi-fis-fluidi.js	33b66d5c555b28b0b65224aef0d23b1f113fdc402ef58e00de4e2c630e38fc7f
bsi-mat-vettoriale.js	ced5e1917ee286455c37ca008c9154b74118ef40021c7df6b97561437c544fdb
bsi-chi-cinetica.js	e1a42d19404deb16f0c9cf9ac19de6b0970c955ec7b4427c4838ca97705bf3b4
bsi-fis-meccanica.js	a73522108a4fdcd7b99719157b87111af8520f219c3a086d6d73eee6fefed569
bsi-fis-onde.js	83ac1416df79f6fde4057382143fce24f9c6bb178f10fd44d44c96c0ffe6660a
bsi-fis-circuiti.js	098679cdbb52241b6035e23076c82a25fe3d9a72b6f5bf7d4f880f887223f375
bsi-fis-termo.js	61a151f01860e31d7c5470d6ace6e6ee8d007c4797f316cf6b005d8de21d8224
bsi-mat-numerici.js	8656bdf657dcc3468f78951ed3d8e1056d953b147e04595c6bfaad647fa8ca93
bsi-mat-geometria.js	609bf6c951d438aa0b0639fd2e736a52b340a363294ee564acfe9510966334be
bsi-chi-equilibrio.js	15d9d99c63c1db49dc3834b7b1db83f5ad2e03d1b9cc1b2f69cdb8ba6cffcefd
bsi-lab3d.js	9b1424f3ad744cc2abd42be04f9db0907578452ecb66156e2c809a5c774a1604
bsi-vista-molecole.js	4a7eb7cd7d8a0033ad0381d936d8e36b2b2e6b75af04988ecc5e9f4e8022cbc4
bsi-assistente.js	103375186ee638e89e324a0d737be266c8464df20d763bb85edafe0e6af4daef
bsi-chi-esame.js	ddb5d73fcf887e03770b3f0794fecaa2f8c918f4fae445bf57ff7a637f4643e7
bsi-fis-elettrodi.js	249bb92ac39b78e916eaa0ecf80a2b1aac5d10249bb6b51628f61839c4b64240
bsi-fis-superfici.js	3daec64d2c4755e70279b8ef2f433e4c98d6e90ff7d8c7d9c0689a24f080abd8
bsi-mat-statistica.js	0b0c82afa13173b84308589cf16fbaa8b30c5e575fd8fdee00034b70c34d17fc
bsi-nmr2d.js	a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78
bsi-moldraw.js	45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e
bsi-documento.js	0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9
bsi-quesito.js	6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf
bsi-spettrolettore.js	14daf41305b0a7271c6ff5f7c9078a4e5f0e0149f32460a9608ed1996f810256

File	SHA-256
`bsi-mol3d.js`	`93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893`
`bsi-lingue.js`	`054821470fe9228c314ab8e9801ff84be1dce30358dc0a52501012d4e8397a34`
`bsi-molingue.js`	`e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb`
`bsi-pannelli-lingua.js`	`9437192c6a3995425b3ef305ab86c70685011d40d789c2cc45736880a7369186`
`sw.js`	`5810a6ffb329157d7f644e129fe8c939f50deea8256ed2a443f0978ca4177090`
`rdkit_lab.html`	`066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad`
`astro.html`	`3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d`
`chimorga.html`	`047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d`
`proxy/spectra-proxy.js`	`808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962`

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
`playwright-core`	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, CC BY 4.0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

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